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Central Limit Theorem for independent and identical distributed variables.

Proof via charecteristic functions.

Contents

1 Introduction.

CLT states that taking sufficiently large sample with a finite expectation and variance, the distribution of means converges to **normal distribution** independently from initial state (distribution). The simpliest example is an infinite coin flip. Monte-Carlo is also based on the principle.

2 Weak convergence.

According to Borovkov (p.156,1999), uniform convergence, i.e. central limit theorem for identical distributed random variables, is a consequence of the weak convergence and continious property of standard normal distribution Φ. Thus, lets start the proof by recalling the theorem.

Definition 1

$F_n(x)$ weakly converges to $F(x)$: $F_n(x) \Rightarrow F(x)$, if for every real and bounded function f, it is satisfied:

$$\int f dF_x(x) \to \int f dF(x) \tag{1}$$

If probability measures P_X and P satisfy:

$$\int_A f dP_x \to \int_A f dP \tag{2}$$

for every real and bounded function f on A, then P_X **converges weakly** to $P : P_X \Rightarrow P$.

It gives an important concept:

$$P_X \Rightarrow P \text{ iif } x \to x_n, \tag{3}$$

where P_X and P - mass functions for x_n and x correspondingly.

Intuitively, it means that if the set of real values weakly converges to continious function, then its probability measures converges, in addition, uniquely. The following definition can be defined for random variables and be enlarged for proof of convergence to Wiener measure (e.g. , functional CLT). See proof in Billingsly(p.21, 1977).

3 Charateristic function.

3.1 Definitions.

- Probability space $(\Omega, \mathcal{F})$ will be used. Define measurable space $(\mathbf{E}, \mathcal{E}) = (\mathbf{Z}, \mathcal{B}(\mathbf{Z}))$, where $\mathbf{Z}$ is a set of complex numbers $z = x + iy$ and $x, y \in \mathbb{R}$; $\mathcal{B}(\mathbf{Z})$ is the smallest $\sigma - algebra$ containing set $\{z : z = x + iy,\ a_1 < x \leq b_1, a_2 < x \leq b_2\}$.

- Let $Z(w) = X(w) + iY(w)$ be complex random variable, where $X = X(\omega)$ and $Y = Y(\omega)$ functions defined on Ω, taking values in $\mathbf{E}$ and $\mathcal{F}/\mathbf{E} - measurable$ (i.e. elements are taken from $\mathbf{E}$).

  Thus, the complex random variable inherits main statistical properties of random variables. For instance: $\mathsf{E}(Z) = \mathsf{E}(X) + i \cdot \mathsf{E}(Y)$ for complex random variable $Z = X + i \cdot Y$.

- Complex random variables $Z_1 = X_1 + i \cdot Y_1$ and $Z_2 = X_2 + i \cdot Y_2$ are independent if (X_1, Y_1) and (X_2, Y_2) are independent.

Definition 2

Let $F = F(x)$ be n-dimensional distribution function in $(\mathcal{R}^n, \mathcal{B}(\mathcal{R}^n))$ and $x = (x_1, x_2, \ldots, x_n)$. Then characteristic function is defined as:

$$\varphi(t) = \int e^{i(t,x)} dF(x) \tag{4}$$

where $t \in \mathcal{R}^n$.

Definition 3

Let $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_n)$ - random vector defined in $\mathcal{R}^n$, then its characteristic function is:

$$\varphi_\gamma(t) = \int e^{i(t,x)} dF_\gamma(x) \tag{5}$$

where $t \in \mathcal{R}^n$, $F_\gamma = F_\gamma(x)$ - a distribution function of the vector.

Hence, at this point characteristic function is defined for vectors of both: sequence of real numbers and random variables. Applying the Change-of-Variable Theorem for Lebesgue Integral, the one of the main property can be derived:

$$\varphi_\gamma(t) = \mathsf{E} e^{i(t,\gamma)}, t \in \mathcal{R}^n \tag{6}$$

The Change-of-Variable Theorem for Lebesgue Integral.

Let $(\Omega, \mathcal{F})$ and $(\mathbf{E}, \mathcal{E})$ be measurable spaces where $X = X(\omega)$ is $\mathcal{F}/\mathcal{E} - measurable$ function with values in $\mathbf{E}$. Let P - probability measure on $(\Omega, \mathcal{F})$ and P_X - probability measure on $(\mathbf{E}, \mathcal{E})$ caused by $X = X(\omega)$:

$$P_X(A) = \mathsf{P}\{\omega : X(\omega) \in A\}, A \in \mathcal{E} \tag{7}$$

Then for any $\mathcal{E} - measurable$ function $g = g(x), x \in \mathbf{E}$,

$$\int_A g(x) P_X(dx) = \int_{X^{-1}(A)} g(X(\omega)) \mathsf{P}(d\omega), A \in \mathcal{E} \tag{8}$$

Therefore: both integrals exist and coincide. As consequence, the (3) can be applied for any random variables vectors.

Example: Let X be a random variable with distribution function $F = F(x)$. Then its characteristic equation can be defined as $\varphi(t) = \mathsf{E}e^{itX}$.

3.2 Key properties.

1. Let X, Y be random variables. If $Y = aX + b$, then:

$$\varphi_Y(t) = \mathsf{E}e^{itY} = \mathsf{E}(e^{it(aX+b)}) = e^{itb}\mathsf{E}(e^{itaX}) = e^{itb}\varphi_X(at)$$

2. If $X_1, X_2, \dots, X_n$ - iid and $S_n = X_1 + X_2 + \dots + X_n$, then:

$$\varphi_{S_n}(t) = \mathsf{E}e^{itS_n} = \mathsf{E}e^{it(X_1+X_2+\dots+X_n)} = \mathsf{E}e^{itX_1} \cdot \mathsf{E}e^{itX_2} \cdot \dots \cdot \mathsf{E}e^{itX_n} = \prod_{j=1}^{n} \varphi_{X_j}(t)$$

3. $\varphi_\gamma(t)$ is uniformly continious function:

$$|\varphi(t+h) - \varphi(t)| = |\mathsf{E}(e^{i(t+h)\gamma} - e^{it\gamma})| \leq \mathsf{E}|e^{ih\gamma - 1}| \to 0$$

4. If k^{th} moment exists $\mathsf{E}|\gamma|^k < \infty$, $k \geq 1$, then continious $\exists\ k^{th}$ derivative of function $\varphi_\gamma(t)$ and $\exists\ \varphi^{(k)}(0) = i^k \mathsf{E}\gamma^k$.

   Proof: As $|\int ixe^{itx}dF(x)| \leq \int |x|dF(x) = \mathsf{E}|\gamma| < \infty$ then derivative of characteristic function $\int ixe^{itx}dF(x)$ converges uniformly relatively to t. Here the fact that $|e^{it}| = 1$ and the Weierstrass Feature of uniform convergence for integrals depending on the parameter are used.

5. The derivative of the characteristic function is:

$$\varphi'(t) = \frac{d}{dt}\int e^{itx}dF(x) = i\int xe^{itx}dF(x) \text{ and if } t = 0 : \varphi'(0) = i\mathsf{E}(X).$$

How to define that function $\varphi(t)$ is indeed characteristic function?

Bochner-Khinchin Theorem

It is necessary and sufficient it to be non-negatively defined: for any real $t_1, t_2, \dots, t_n$ and any complex numbers $\lambda_1, \dots, \lambda_n$:

$$\sum_{k,j=1}^{n} \varphi(t_k - t_j)\lambda_k\bar{\lambda}_j \geq 0, \ \varphi(0) = 1$$

Using the fact that $\varphi(t) = \mathsf{E}e^{it\gamma}$, obtain:

$$\sum_{k,j=1}^{n} \varphi(t_k - t_j)\lambda_k\bar{\lambda}_j = \mathsf{E}\sum_{k,j=1}^{n} e^{i(t_k - t_j)\gamma}\lambda_k\bar{\lambda}_j = \mathsf{E}|\sum_{k=1}^{n} \lambda_k e^{it_k\gamma}|^2 \geq 0$$

Note: Conjecture of complex numbers give the following property:

Denote $\lambda = a + ib$, then $\bar{\lambda} = a - ib$.

Then:

$$\lambda\bar{\lambda} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 + b^2 \geq 0$$

And: $\Re = a^2 + b^2, \Im = 0$.

Hence, the matrix of characteristic function defined for some real $t_1, t_2, \dots, t_n$ and any complex numbers $\lambda_1, \dots, \lambda_n$ is positive definite and can be produced by probability measures. (It is somehow important for stationary process with positive correlation function).

3.3 Characteristic function for Normal distribution (take as given).

Let $\xi \sim N(0, 1)$, then

$$\phi(t) = \phi_\xi(t) = \frac{1}{\sqrt{2\pi}} \int e^{itx - x^2/2} dx \tag{9}$$

Taking the derivative:

$$\phi'(t) = \frac{1}{\sqrt{2\pi}} \int ixe^{itx - x^2/2} dx = -\frac{1}{\sqrt{2\pi}} \int te^{itx - x^2/2} dx = -t\phi(t) \tag{10}$$

We obtain differential equation $\phi'(t) = -t\phi(t)$. Solving it and taking into account that $\phi(0) = 1$, obtain $c = 0$ and $\phi(t) = e^{-t^2/2}$.

4 The continuity (convergence) theorem.

Theorem

For weak convergence $F_n \Rightarrow F$ it is necessary and sufficient that $\varphi_n(t) \to \varphi(t)$ for any t, where $\varphi(t)$ - charateristic function of F.

5 CLT proof.

Suppose $\xi_1, \xi_2, \ldots, \xi_n$ - iid, $\mathsf{E}(\xi_i) = \mu$, $\mathsf{Var}(\xi_i) = \sigma^2$.
Then $S_n = \xi_1 + \xi_2 + \ldots + \xi_n$, $\mathsf{E}(S_n) = \mu n$ and $\mathsf{Var}(S_n) = \sigma^2 n$.
Let $\zeta_n = \frac{S_n - \mathsf{E}(S_n)}{\sqrt{\mathsf{Var}(S_n)}} = \frac{S_n - \mu n}{\sigma\sqrt{n}}$.

Theorem (Borovkov,1977)

If $0 < \sigma^2 < \infty$, then $P(\zeta_n < x) \to \Phi(x)$ as $n \to \infty$ uniformly relative to x $(-\infty < x < \infty)$.

Therefore: the sequence $\{\zeta_n\}$ is asymptotically normal.

Note:

If sequence $\{\xi'_n\}_{n\geq 0}$ s.t. $\xi'_i = \xi_i - \mu$ is considered, then $S'_n = \sum \xi'_i = \sum(\xi_i - \mu) = S_n - n\mu$.
Then $\mathsf{E}(S'_n) = \mathsf{E}(S_n - n\mu) = 0$, $\mathsf{Var}(S'_n) = \mathsf{Var}(S_n) = \sigma^2 n$.
Obtain: $\zeta'_n = \frac{S'_n - 0}{\sqrt{\mathsf{Var}(S'_n)}} = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \zeta_n$ for any μ. Thus, we can take $\mu = 0$.

The characteristic function for ζ_n:

$$\varphi_{\zeta_n} = \mathsf{E}e^{it\zeta_n} = \mathsf{E}e^{it(\frac{S_n - n\mu}{\sigma\sqrt{n}})} = \mathsf{E}e^{it\frac{S_n}{\sigma\sqrt{n}}} = \mathsf{E}e^{it(\frac{\xi_1+\xi_2+\ldots+\xi_n}{\sigma\sqrt{n}})} = \mathsf{E}e^{\frac{it}{\sigma\sqrt{n}}\xi_1}\mathsf{E}e^{\frac{it}{\sigma\sqrt{n}}\xi_2} \cdot \ldots \cdot \mathsf{E}e^{\frac{it}{\sigma\sqrt{n}}\xi_n} = \varphi^n(\frac{t}{\sigma\sqrt{n}}) \tag{11}$$

Now take Taylor expansion for $\varphi(x)$, where $x = \frac{t}{\sigma\sqrt{n}}$ at x = 0; $\mathsf{E}(\gamma) = 0, \mathsf{Var}(\gamma) = \sigma^2$:

$$\varphi(x) = \varphi(0) + \varphi'(0)x + \varphi''(0)\frac{x^2}{2} + o(x^2) = 1 + ix\mathsf{E}(\gamma) - \frac{x^2}{2}\mathsf{E}(\gamma^2) + o(x^2) \tag{12}$$

1. $\varphi(0) = \mathsf{E}e^{ix\gamma} = \mathsf{E}e^{i\frac{t}{\sigma\sqrt{n}}\gamma} = 1;$
2. $\varphi'(0) = i\mathsf{E}(\gamma) = 0;$
3. $\varphi''(0) = i^2\mathsf{E}(\gamma^2) = -\sigma^2$

Plugging into equation, obtain:

$$\varphi(x) = \varphi(\frac{t}{\sigma\sqrt{n}}) = 1 - \frac{\sigma^2}{2}(\frac{t}{\sigma\sqrt{n}})^2 + o(\frac{t^2}{n}) = 1 - \frac{t^2}{2n} + o(\frac{t^2}{n}) \tag{13}$$

And:

$$\varphi_{\zeta_n} = \varphi^n(\frac{t}{\sigma\sqrt{n}}) = (1 - \frac{t^2}{2n})^n \to e^{-\frac{t^2}{2}} \text{ as } n \to \infty \tag{14}$$

Hence: $\zeta_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \to Norm(0,1)$ as $n \to \infty$, *QED*.